

Max Tong

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EDUCATION

Northwestern University

Expected Jun 2027

Bachelor of Science in Computer Science - Minor in Math - Major GPA: 3.7/4.0

Evanston, IL

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (C/C++), Artificial Intelligence, Machine Learning, Scalable Software (AWS), Networking (TCP/UDP)
- **Clubs & Organizations:** Northwestern Garage Residency, Formula SAE, Competitive Programming (ICPC)

WORK EXPERIENCE

AT&T

Atlanta, GA

Software Engineering Intern

Jun 2026 - Present

- Engineered **Java/FastAPI REST microservices** with **Redis** caching powering real-time fraud detection for Cricket Wireless, screening **10,000+** daily transactions at **sub-250ms p95** latency
- Designed **event-driven** detection pipelines that shifted fraud scoring from delayed batch review to real-time inline checks, cutting detection time for account-takeover, payment, and signup fraud from hours to milliseconds
- Cut scoring latency by **40%** under production-scale traffic by optimizing **MongoDB** schema design, indexing, and query execution, enabling inline fraud checks on every transaction without impacting legitimate users

TwoCann

Evanston, IL

Founder

Jun 2024 - Present

- Built and deployed cloud-backed app with **Firebase, Google Cloud Platform**, serving **1,000+** registered users.
- Wrote native **Swift** code for iOS widgets alongside **Dart** and **JavaScript**; designed efficient data models and caching to reduce Firestore reads/writes by **30%** and integrated Firebase FCM tokens for push notifications.
- Built and debugged web service (HTTPS callable functions), utilized GCP logging/metrics to fix reliability issues.
- Led a team in Northwestern's startup incubator (The Garage); shipped features using Git version control.

Clemson University

Clemson, SC

Research Assistant

Jun 2025 - Sep 2025

- Developed a serverless web app with **React/Node.js** for user check-in for lab machinery. Used **AWS RDS** to track user sessions and machine usage across shared equipment & **S3** to store lab setting snapshots & survey data
- Implemented a **CI/CD** pipeline using GitHub Actions to automatically build, test, and deploy after every commit
- Built and deployed a Hugo static website used by **200+** internal users to access documentation and lab resources.
- Debugged cross platform runtime issues in a **Linux Docker** container, identifying several OS-dependent bugs.

RESEARCH

Prescience Lab | C/C++, RDMA, Networking

Dec 2025 - Present

- Developed an RDMA gather function in C using a novel triggered collective architecture, lowering MPI latency.
- Collaborated with PhD students to engineer a kernel bypass networking flow for direct NIC communication between machines in super computing clusters. Potential implications for ML and HFT.
- Investigated floating point attack vectors & interactions with chaos theory to boost system and simulation security.

PROJECTS

Reliable Transport Protocol from Scratch | Python, Concurrency, Networking

Dec 2025 - March 2026

- Engineered a copy of TCP, simulating packet loss, data corruption, and ensuring reliable delivery through sequence numbering, MD5 packet hashing, and timeout based packet retransmission.
- Maximized network throughput by implementing pipelined ACKs and background listener threads.

Test Driven Monopoly Game | Java, JUnit, EasyMock,

March 2026 - Jun 2026

- Developed a full Monopoly game in Java following TDD, writing unit and integration tests for every function.
- Engaged in manual and automated testing techniques, achieving 100% basis path and mutation coverage generated from Jacoco & PITest.

TECHNICAL SKILLS

Languages: C/C++, Rust, Python, Java, JavaScript/TypeScript, SQL (Postgres), Swift, HTML/CSS, Go, MatLab

Frameworks & Stack: React, Express, Flutter, Flask, PyTorch, TensorFlow, AWS, Kubernetes, Firebase, Docker

Systems & Tools: Linux, Git, CI/CD, Testing (Google Test), Data Structures & Algorithms, Backend systems/APIs